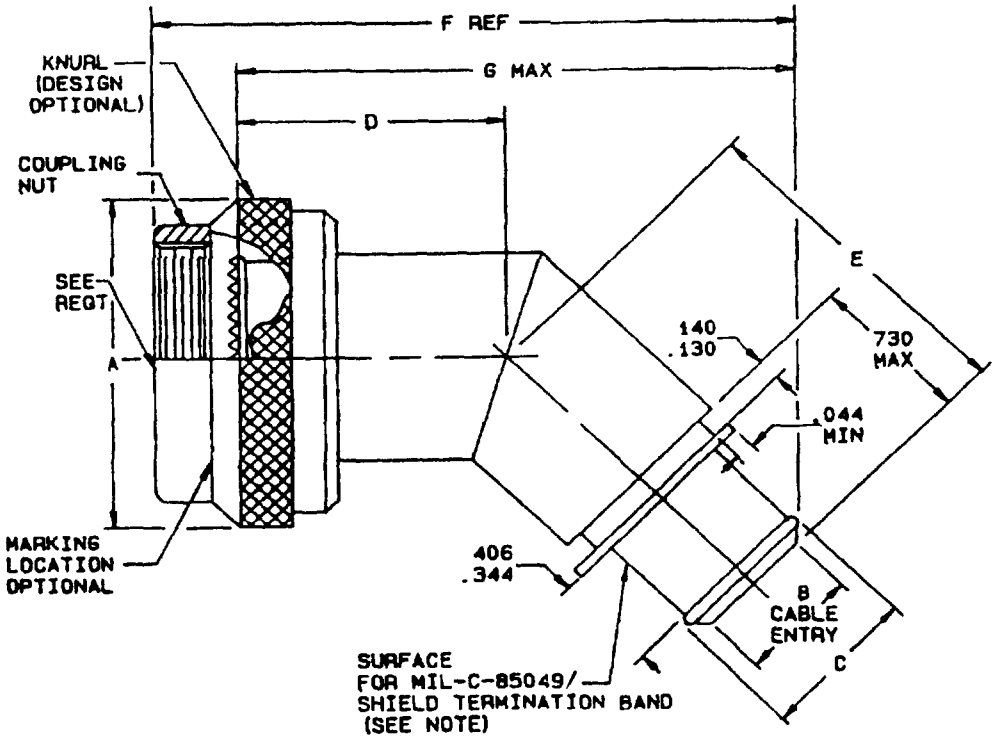
MILITARY SPECIFICATION SHEET

CONNECTOR ACCESSORIES, ELECTRICAL, BACKSHELL, 45°, SELF-LOCKING, SHIELD BAND TERMINATION, (RFI/EMI), SHRINK SLEEVE ACCOMMODATION, CATEGORY 3B (FOR MIL-C-38999 SERIES III AND IV CONNECTORS)

This specification is approved for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation. MIL-C-85049

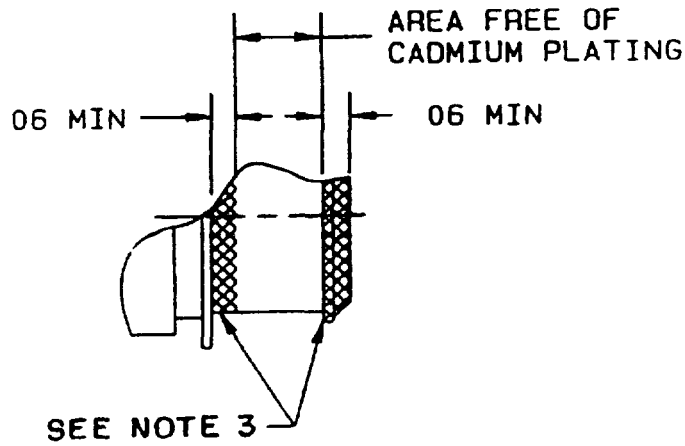


NOTE Ribbed, internal knurl or surface roughness R_a 125-250 in accordance with ANSI B46 1 See figure 2 for selective plating details

FIGURE 1 Configuration and dimensions

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Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
044	1 12	500	12 70	770	19 56	1 010	25 65	1 157	29 39	1 279	32 49
060	1 52	562	14 27	812	20 62	1 030	26 16	1 160	29 46	1 300	33 02
130	3 30	583	14 81	829	21 05	1 060	26 92	1 170	29 72	1 330	33 78
140	3 56	625	15 88	858	21 79	1 080	27 43	1 190	30 23	1 350	34 29
250	6 35	645	16 38	895	22 73	1 083	27 51	1 200	30 48	1 406	35 71
312	7 92	688	17 47	938	23 83	1 110	28 19	1 210	30 73	1 516	38 51
395	10.03	707	17 96	957	24 31	1 120	28 45	1 240	31 50	1 642	41 71
438	11 13	730	18 52	984	24 99	1 145	29 08	1 260	32 00	1 768	44 91
457	11 61	750	19 05	1 000	25 40	1 150	29 21	1 270	32 26	1 889	47 98

NOTES
1. Dimensions are in inches
2. Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm
3. Cadmium/nickel interface shall be coated with polysulfide sealant

FIGURE 2 Selective plating

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TABLE I Shell size and dimensions

Accessory shell size	Shell size code (ref)	A max dia	B dia ± 010 entry size		C dia ref		D max dim	E max dim	G max C dia		F ref C dia	
			02	03	02	03			02	03	02	03
9	A	858	N/A	250	N/A	395	1 01	1 16	N/A	1 970	N/A	2.218
11	B	984	N/A	312	N/A	457	1 03	1 19	N/A	2 033	N/A	2.281
13	C	1 157	312	438	457	583	1 06	1 21	2.077	2 122	2 325	2.370
15	D	1.279	438	562	583	707	1 08	1 24	2 163	2 207	2 411	2 455
17	E	1 406	500	625	645	770	1 11	1 26	2 229	2.273	2 477	2.521
19	F	1 516	625	750	770	895	1 12	1 27	2 290	2 334	2 538	2.528
21	G	1 642	.625	812	770	957	1 15	1 30	2 341	2.408	2.589	2.656
23	H	1 768	688	938	829	1 083	1.17	1 33	2.404	2 493	2 652	2.741
25	J	1 889	750	1.000	895	1 145	1 20	1 35	2 471	2 559	2.719	2.807

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REQUIREMENTS

Design and construction

Dimensions and configuration See figure 1 and table I

Interface dimensions Shall conform to MIL-C-85049

Coupling nut Shall be captivated to the adapter body and free to rotate

Material and finish In accordance with MIL-C-85049 and table II herein

TABLE II Material and finish

Finish	Material
G - Electroless nickel 1/ (space grade) N - Electroless nickel 1/ P - Cadmium olive drab over electroless nickel (selective plating, see figure 2) W - Cadmium olive drab	Aluminum alloy
J - Cadmium olive drab L - Cadmium olive drab over electroless nickel (selective plating, see figure 2) 2/ M - Electroless nickel	Composite

1/ Not for Navy use Limited to space applications for Air Force use
2/ Must meet the requirements, and pass tests for finish J

Vibration Backshells shall be subjected to test parameters of applicable connector specifications The coupling torque shall be within +20, -10 inch-pounds of the initial value

Additional tests for finishes J, L, and M shall be as follows

Ozone exposure Backshells shall be tested in accordance with MIL-STD-1344, method 1007

Sample Each sample shall consist of one backshell

Sample preparation Not applicable

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Failure Blistering, or pealing of the plating constitutes a failure Any condition which adversely affects the function of the backshell also constitutes a failure.

Fluid immersion (see table III). Backshell shall be tested in accordance with MIL-STD-1344, method 1016

Sample: Each sample shall consist of one backshell.

After immersion backshells shall meet the coupling thread strenght requirements of MIL-C-85049

TABLE III Test fluids

Sample number	Test fluid (method 1016 of MIL-STD-1344)
1	d
2	e
3	a
4	b
5	f
6	g
7	c
8	l
9	h
10	i
11	j
12	k

Thermal vacuum outgassing (finish G) Backshell shall not emit vacuum condensable, noxious or toxic gasses when tested in accordance with ASTM E595 Total mass loss (TML) shall not be greater than 1.0 percent and collected volatile condensable materials (CVCM) shall not be greater than 10 percent Data listed in NASA Reference Publication 1124 Revised may be used in lieu of actual test data for application materials

Part or Identifying Number (PIN) See example below

M85049/ 89 - 15 P 02

Military designator

Specification sheet

Shell size, see table I

Finish

Entry size

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QUALIFICATION:

Qualification See MIL-C-85049, category 3B

QPL evaluating activity Naval Air Warfare Center, Code 8/444, 6000 East 21st Street, Indianapolis, IN 46219-2189

CONCLUDING MATERIAL

Custodians
Army - CR
Navy - AS
Air Force - 85

Review activities
Army - AR, MI
Navy - EC, MC, OS
Air Force - 11, 15, 19, 99
DLA - ES

Preparing activity
Air Force - 85

Agent
DLA - ES

(Project 5935-3760-08)

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CUSTOMER SATISFACTION INQUIRY

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FAX - COM 513-296-8783 DSN 986-8783

VERY DISSATISFIED	DISSATISFIED	SOMEWHAT DISSATISFIED	SOMEWHAT SATISFIED	SATISFIED	VERY SATISFIED	NOT APPLICABLE
1	2	3	4	5	6	7

(PLEASE ENTER THE VALUE THAT YOU FEEL IS APPROPRIATE)

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III. COMMENTS/SUGGESTIONS

(CONSTRUCTIVE COMMENTS OR SUGGESTIONS APPRECIATED)

MIL-C-85049/89

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